

MATH 3260 - Chapter 2 Worksheets

Matrices

Name: _____

Date: _____

These are student practice worksheets. They are designed to be used with the chapter notebook and notes, but every problem below contains the data needed to work it independently. No solutions are included in this student version.

Section 2.1 - Operations with Matrices

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Determine the size of each matrix and identify the requested entries:

$$A = \begin{bmatrix} 9 & 20 & 21 \\ 1 & 5 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 8 \\ 12 \end{bmatrix}.$$

Find the sizes of A and B , a_{23} , and b_{31} .

2. Find p and q so that

$$\begin{bmatrix} p & 9 \\ 20 & q \end{bmatrix} = \begin{bmatrix} 1 & 9 \\ 20 & 21 \end{bmatrix}.$$

3. Given

$$C = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 2 \\ -1 & 6 \end{bmatrix},$$

find $C + D$, $C - D$, and $3C - 2D$.

4. Determine whether each product is defined and, if so, state its size:

(a) $(3 \times 4)(4 \times 2)$

(b) $(2 \times 3)(2 \times 2)$

(c) $(1 \times 5)(5 \times 1)$

5. Compute

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 0 \\ 3 & 2 \end{bmatrix}.$$

For the $(1, 1)$ entry, explicitly identify the row and column whose dot product creates the entry.

6. Write the system as $A\mathbf{x} = \mathbf{b}$ and then express $\mathbf{b}$ as a linear combination of the columns of A :

$$\begin{cases} x + 2y - z = 9, \\ 2x + y + 3z = 20. \end{cases}$$

Section 2.2 - Properties of Matrix Operations

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Solve $4X - A = B$ for X , where

$$A = \begin{bmatrix} 4 & 8 \\ 12 & 16 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 0 \\ 4 & 20 \end{bmatrix}.$$

2. Verify $c(A + B) = cA + cB$ for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & -1 \\ 2 & 0 \end{bmatrix}, \quad c = 3.$$

3. Compute AB and BA for

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}.$$

What does the comparison show about commutativity?

4. Verify associativity, $(AB)C = A(BC)$, for

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}.$$

5. Find M^T and $(M^T)^T$ for

$$M = \begin{bmatrix} 1 & 9 & 20 \\ 21 & 0 & 5 \end{bmatrix}.$$

6. Verify $(AB)^T = B^T A^T$ for the matrices in Problem 3.

7. Verify that

$$A = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

are both nonzero but $AB = O$. Explain why the scalar zero-product rule does not carry over to matrices.

Section 2.3 - The Inverse of a Matrix

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Find the inverse of

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

using the 2×2 inverse formula. Verify both $AA^{-1} = I$ and $A^{-1}A = I$.

2. Determine whether

$$B = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}$$

is singular or nonsingular, and state whether it is invertible.

3. Find the inverse of

$$C = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

by Gauss-Jordan elimination. Show every row operation on $[C \mid I]$.

4. Solve

$$\begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

using an inverse.

5. Let

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Find $(PQ)^{-1}$ in two ways and verify $(PQ)^{-1} = Q^{-1}P^{-1}$.

6. Multiple right-hand sides can be solved at once. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 \\ 3 & 0 \end{bmatrix}.$$

Solve $AX = B$ by computing $X = A^{-1}B$. Explain how the two columns of B represent two different systems with the same coefficient matrix.

Section 2.4 - Elementary Matrices and LU-Factorization

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Construct the 3×3 elementary matrix that performs each operation:

(a) $R_1 \leftrightarrow R_3$

(b) $R_2 \leftarrow -2R_2$

(c) $R_3 \leftarrow R_3 + 4R_1$

2. Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 4 & 7 \end{bmatrix}.$$

Construct an elementary matrix E such that EA performs $R_3 \leftarrow R_3 - 4R_1$, then verify the multiplication.

3. Factor

$$B = \begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix}$$

as a product of elementary matrices. Explain which row operation each elementary factor represents.

4. Find an LU-factorization of

$$C = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 3 & 1 \\ -2 & 2 & 4 \end{bmatrix}.$$

As you eliminate, record each multiplier and show where it is stored in L .

5. Using your factorization from Problem 4, solve

$$C\mathbf{x} = \begin{bmatrix} 4 \\ 11 \\ 14 \end{bmatrix}.$$

Show the two stages explicitly:

$$L\mathbf{y} = \mathbf{b}, \quad U\mathbf{x} = \mathbf{y}.$$

Section 2.5 - Markov Chains

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

Convention for this course: $p_{ij} = P(\text{next state } i \mid \text{current state } j)$. Columns are starting states, rows are next states, and every column of a transition matrix sums to 1.

1. Determine whether each matrix is column-stochastic:

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}, \quad B = \begin{bmatrix} 0.6 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}.$$

For A , interpret a_{11} and a_{21} in words using “start in State 1” and “next state.”

2. For

$$P = \begin{bmatrix} 0.75 & 0.25 \\ 0.25 & 0.75 \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix},$$

find $\mathbf{x}_1$, $\mathbf{x}_2$, and $\mathbf{x}_5$. State what $\mathbf{x}_1$ means.

3. Find the steady state of the matrix in Problem 2 and normalize it so its entries sum to 1.

4. For

$$P = \begin{bmatrix} 0.6 & 0.2 & 0.1 \\ 0.3 & 0.7 & 0.2 \\ 0.1 & 0.1 & 0.7 \end{bmatrix},$$

verify that P is column-stochastic and find the next state from

$$\mathbf{x}_0 = \begin{bmatrix} 0.4 \\ 0.4 \\ 0.2 \end{bmatrix}.$$

Show at least one component calculation explicitly.

5. Consider the absorbing chain

$$P = \begin{bmatrix} 0.4 & 0 & 0 \\ 0.35 & 1 & 0 \\ 0.25 & 0 & 1 \end{bmatrix}.$$

Identify the absorbing states and find the limiting absorption probabilities when the process starts in State

1. Explain what “absorbing” means in the context of a terminal-outcome model.

Section 2.6 - More Applications of Matrix Operations

Show enough work to make your reasoning clear. Interpret your answer when a problem asks for interpretation.

1. Use the mapping blank = 0, $A = 1, \dots, Z = 26$ and

$$E = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

to encode the message **RNB MUSIC**. Pad with a blank if necessary.

2. Find E^{-1} and decode the first two coded blocks from Problem 1.

3. A fictional two-industry economy has

$$D = \begin{bmatrix} 0.20 & 0.10 \\ 0.25 & 0.20 \end{bmatrix}, \quad \mathbf{e} = \begin{bmatrix} 70 \\ 55 \end{bmatrix}.$$

Solve $(I - D)\mathbf{x} = \mathbf{e}$. Then verify one component by separating internal use from external demand.

4. Find the least-squares line for

$$(1, 4), (2, 7), (3, 8), (4, 13).$$

Show X , $X^T X$, $X^T \mathbf{y}$, the normal equations, the fitted line, and the SSE.